

Mock Proof Questions and Solutions

Multiple Linear Regression and Polynomial Regression

Focus: ordinary derivatives, matrix transformations, gradients, and substitution.

Notation and derivative rules

For the MLR model, let

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{y} \in \mathbb{R}^{n \times 1}, \quad \mathbf{X} \in \mathbb{R}^{n \times p}, \quad \boldsymbol{\beta} \in \mathbb{R}^{p \times 1}.$$

The i th row of $\mathbf{X}$ is

$$\mathbf{x}_i^\top = (1, x_{i1}, \dots, x_{i,p-1}).$$

The most useful matrix identities are

$$(AB)^\top = B^\top A^\top, \quad (A+B)^\top = A^\top + B^\top, \quad (A^{-1})^\top = (A^\top)^{-1},$$

and

$$\mathbf{X}^\top \mathbf{X} \text{ is symmetric because } (\mathbf{X}^\top \mathbf{X})^\top = \mathbf{X}^\top \mathbf{X}.$$

For a vector $\boldsymbol{\beta}$ and a constant vector a ,

$$\nabla_{\boldsymbol{\beta}}(a^\top \boldsymbol{\beta}) = a.$$

For a constant matrix A ,

$$\nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\top A \boldsymbol{\beta}) = (A + A^\top)\boldsymbol{\beta}.$$

If A is symmetric, this becomes

$$\nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\top A \boldsymbol{\beta}) = 2A\boldsymbol{\beta}.$$

Question 1 — Derive the MLR OLS estimator

Consider the model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

Define the ordinary least-squares loss

$$L(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

- (a) Expand $L(\boldsymbol{\beta})$.
- (b) Compute the gradient $\nabla_{\boldsymbol{\beta}} L$ clearly.
- (c) Set the gradient equal to zero and derive the normal equations.
- (d) Assuming $\mathbf{X}^\top \mathbf{X}$ is invertible, derive the OLS estimator.

Solution

First expand the loss:

$$\begin{aligned} L(\boldsymbol{\beta}) &= (\mathbf{y}^\top - \boldsymbol{\beta}^\top \mathbf{X}^\top)(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \\ &= \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}. \end{aligned}$$

The two middle terms are scalars, and

$$\mathbf{y}^\top \mathbf{X}\boldsymbol{\beta} = (\mathbf{y}^\top \mathbf{X}\boldsymbol{\beta})^\top = \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y}.$$

Therefore,

$$L(\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Now differentiate term by term:

$$\nabla_{\boldsymbol{\beta}}(\mathbf{y}^\top \mathbf{y}) = 0,$$

$$\nabla_{\boldsymbol{\beta}}(-2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y}) = -2\mathbf{X}^\top \mathbf{y},$$

and, because $\mathbf{X}^\top \mathbf{X}$ is symmetric,

$$\nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}) = 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Hence,

$$\boxed{\nabla_{\boldsymbol{\beta}} L = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}}.$$

At the minimizer $\hat{\boldsymbol{\beta}}$, the gradient is zero:

$$-2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = 0.$$

Divide by 2:

$$\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}.$$

These are the normal equations. Premultiplying by $(\mathbf{X}^\top \mathbf{X})^{-1}$ gives

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}}.$$

Dimension check:

$$(p \times n)(n \times n)(n \times 1) = p \times 1,$$

so the result has the correct dimension for $\hat{\boldsymbol{\beta}}$.

Question 2 — Derive the MLR normal equations using ordinary derivatives

For

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{p-1} x_{i,p-1} + \varepsilon_i,$$

define

$$L(\beta_0, \dots, \beta_{p-1}) = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^{p-1} \beta_j x_{ij} \right)^2.$$

Show using only ordinary partial derivatives that the normal equations are

$$\sum_{i=1}^n x_{ij} \left(y_i - \sum_{k=0}^{p-1} x_{ik} \beta_k \right) = 0, \quad j = 0, 1, \dots, p-1,$$

where $x_{i0} = 1$. Then show that these equations are exactly

$$\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^\top \mathbf{y}.$$

Solution

Let

$$r_i = y_i - \sum_{k=0}^{p-1} x_{ik} \beta_k.$$

For a fixed coefficient β_j ,

$$\frac{\partial r_i}{\partial \beta_j} = -x_{ij}.$$

Using the chain rule,

$$\begin{aligned} \frac{\partial L}{\partial \beta_j} &= \sum_{i=1}^n 2r_i \frac{\partial r_i}{\partial \beta_j} \\ &= -2 \sum_{i=1}^n x_{ij} r_i. \end{aligned}$$

Set this equal to zero:

$$\sum_{i=1}^n x_{ij} \left(y_i - \sum_{k=0}^{p-1} x_{ik} \beta_k \right) = 0.$$

Rearrange:

$$\sum_{k=0}^{p-1} \left(\sum_{i=1}^n x_{ij} x_{ik} \right) \beta_k = \sum_{i=1}^n x_{ij} y_i.$$

The (j, k) entry of $\mathbf{X}^\top \mathbf{X}$ is

$$(\mathbf{X}^\top \mathbf{X})_{jk} = \sum_{i=1}^n x_{ij} x_{ik},$$

and the j th entry of $\mathbf{X}^\top \mathbf{y}$ is

$$(\mathbf{X}^\top \mathbf{y})_j = \sum_{i=1}^n x_{ij} y_i.$$

Thus all p scalar equations combine into

$$\boxed{\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^\top \mathbf{y}}.$$

Question 3 — Prove that the OLS estimator is unbiased

Assume

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad E(\boldsymbol{\varepsilon}) = \mathbf{0},$$

and

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Prove that

$$E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}.$$

Solution

Substitute the model into the estimator:

$$\begin{aligned}\hat{\boldsymbol{\beta}} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon} \\ &= \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}.\end{aligned}$$

Taking expectations and treating $\mathbf{X}$ and $\boldsymbol{\beta}$ as fixed:

$$\begin{aligned}E(\hat{\boldsymbol{\beta}}) &= E\left[\boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}\right] \\ &= \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top E(\boldsymbol{\varepsilon}) \\ &= \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{0} \\ &= \boxed{\boldsymbol{\beta}}.\end{aligned}$$

Thus OLS is unbiased: across repeated samples, the average estimator equals the true coefficient vector.

Question 4 — Residual orthogonality

Let

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}, \quad \hat{\mathbf{e}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Use the derivatives of the OLS loss to prove

$$\boxed{\mathbf{X}^\top \hat{\mathbf{e}} = \mathbf{0}}.$$

Interpret this result for the intercept and for each predictor.

Solution

The gradient is

$$\nabla_{\boldsymbol{\beta}} L = -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

At $\hat{\boldsymbol{\beta}}$, the gradient is zero:

$$-2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}.$$

Since $\hat{\mathbf{e}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$,

$$\boxed{\mathbf{X}^\top \hat{\mathbf{e}} = \mathbf{0}}.$$

If the first column of $\mathbf{X}$ is a column of ones, then

$$\sum_{i=1}^n \hat{e}_i = 0.$$

For predictor j ,

$$\sum_{i=1}^n x_{ij} \hat{e}_i = 0.$$

Thus the residuals are orthogonal to the intercept column and to every predictor column.

Question 5 — Quadratic polynomial regression

Consider

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i.$$

The loss is

$$L(\beta_0, \beta_1, \beta_2) = \sum_{i=1}^n \left(y_i - \beta_0 - \beta_1 x_i - \beta_2 x_i^2 \right)^2.$$

- (a) Derive all three normal equations using ordinary partial derivatives.
- (b) Solve the first equation for β_0 .
- (c) After substitution, express the remaining two equations using centered transformed predictors.

Solution

Let

$$r_i = y_i - \beta_0 - \beta_1 x_i - \beta_2 x_i^2.$$

The three derivatives are

$$\frac{\partial L}{\partial \beta_0} = -2 \sum_i r_i, \quad \frac{\partial L}{\partial \beta_1} = -2 \sum_i x_i r_i, \quad \frac{\partial L}{\partial \beta_2} = -2 \sum_i x_i^2 r_i.$$

Setting them to zero gives

$$\begin{aligned} \sum_i y_i - n\beta_0 - \beta_1 \sum_i x_i - \beta_2 \sum_i x_i^2 &= 0, \\ \sum_i x_i y_i - \beta_0 \sum_i x_i - \beta_1 \sum_i x_i^2 - \beta_2 \sum_i x_i^3 &= 0, \\ \sum_i x_i^2 y_i - \beta_0 \sum_i x_i^2 - \beta_1 \sum_i x_i^3 - \beta_2 \sum_i x_i^4 &= 0. \end{aligned}$$

Equivalently,

$$\begin{aligned} n\beta_0 + \beta_1 \sum_i x_i + \beta_2 \sum_i x_i^2 &= \sum_i y_i, \\ \beta_0 \sum_i x_i + \beta_1 \sum_i x_i^2 + \beta_2 \sum_i x_i^3 &= \sum_i x_i y_i, \\ \beta_0 \sum_i x_i^2 + \beta_1 \sum_i x_i^3 + \beta_2 \sum_i x_i^4 &= \sum_i x_i^2 y_i. \end{aligned}$$

From the first equation,

$$\beta_0 = \bar{y} - \beta_1 \bar{x} - \beta_2 \bar{x}^2, \quad \bar{x}^2 = \frac{1}{n} \sum_i x_i^2.$$

For the substitution step, define

$$u_i = x_i - \bar{x}, \quad v_i = x_i^2 - \bar{x}^2, \quad w_i = y_i - \bar{y}.$$

Because $\sum_i u_i = \sum_i v_i = \sum_i w_i = 0$, substituting the intercept into the second and third normal equations gives

$$\sum_i u_i^2 \beta_1 + \sum_i u_i v_i \beta_2 = \sum_i u_i w_i,$$

$$\sum_i u_i v_i \beta_1 + \sum_i v_i^2 \beta_2 = \sum_i v_i w_i.$$

Define

$$A_{11} = \sum_i u_i^2, \quad A_{12} = \sum_i u_i v_i, \quad A_{22} = \sum_i v_i^2,$$

$$B_1 = \sum_i u_i w_i, \quad B_2 = \sum_i v_i w_i.$$

Then

$$A_{11}\beta_1 + A_{12}\beta_2 = B_1, \quad A_{12}\beta_1 + A_{22}\beta_2 = B_2.$$

If $D = A_{11}A_{22} - A_{12}^2 \neq 0$, then

$$\boxed{\hat{\beta}_1 = \frac{B_1 A_{22} - B_2 A_{12}}{D}}, \quad \boxed{\hat{\beta}_2 = \frac{B_2 A_{11} - B_1 A_{12}}{D}},$$

and

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} - \hat{\beta}_2 \bar{x}^2}.$$

Question 6 — Polynomial regression as MLR

For the degree- d polynomial model

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i,$$

- (a) Construct the design matrix $\mathbf{X}$.
- (b) Show that the model can be written as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$.
- (c) Derive the OLS estimator from the matrix loss.

Solution

The coefficient vector is

$$\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_d)^\top.$$

The design matrix is

$$\mathbf{X} = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^d \\ 1 & x_2 & x_2^2 & \cdots & x_2^d \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^d \end{pmatrix}.$$

Its i th row times $\boldsymbol{\beta}$ is

$$\begin{pmatrix} 1 & x_i & x_i^2 & \cdots & x_i^d \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_d \end{pmatrix} = \beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d.$$

Thus the polynomial model is an MLR model with transformed predictors

$$x_i, x_i^2, \dots, x_i^d.$$

Using

$$L(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}),$$

the same expansion and gradient calculation as in Question 1 gives

$$\nabla_{\boldsymbol{\beta}} L = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Set this equal to zero:

$$\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}.$$

Therefore,

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}}.$$

Polynomial regression is nonlinear in x , but it is linear in the unknown parameters $\beta_0, \dots, \beta_d$.

Exam checklist

1. Define the residual clearly.
2. Differentiate with respect to each unknown coefficient.

3. Set each derivative equal to zero.
4. For matrix proofs, remember $(AB)^\top = B^\top A^\top$.
5. Combine scalar equations into $\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$.
6. Solve the intercept equation first when substitution is requested.
7. Check dimensions before multiplying matrices.
8. In polynomial regression, use the transformed columns $1, x, x^2, \dots, x^d$.